

Amazon Bedrock Guardrails

What they are, what each policy does, and where they stop

Saturday 22 August 2026 · 11:00–13:00 EAT · Google Meet

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What it is

Guardrails is a configurable safety layer that sits between your users and the model, evaluating both the prompt going in and the response coming out. It is not part of the model. It is an independent policy engine you attach to an inference call.

That independence is the whole design idea. The same guardrail can front an Amazon-hosted model, a self-hosted model, or a third-party model such as OpenAI or Gemini, because the **ApplyGuardrail** API lets you evaluate content without invoking a model at all. Your safety policy stops being a property of whichever model you happened to pick and becomes a property of your application.

Each of the six policies below is optional and independently billed. You enable only what you need.

The six policy types

Policy	What it catches	Typical use
Content filters	Hate, insults, sexual, violence, misconduct and prompt attack — each with its own configurable strength	Baseline moderation for any public-facing application
Denied topics	Topics you define in plain natural language rather than by keyword	"No investment advice." "Do not discuss competitors."
Word filters	A managed profanity list plus your own custom terms	Brand-safety terms, internal codenames
Sensitive information filters	Predefined PII types plus custom regular expressions	Redacting ID numbers, phones and emails from transcripts
Contextual grounding checks	Responses unsupported by retrieved source material, or irrelevant to the question asked	Hallucination control in RAG pipelines
Automated Reasoning checks	Responses that contradict formal rules extracted from a policy document	Regulated decisions where a plausible answer can still be wrong

Two details worth knowing

Sensitive information filters have two modes. Block rejects the request outright. Mask replaces the value with a token such as [NAME-1] or [EMAIL-1] and lets the request through. Both modes now work on input prompts as well as model responses, meaning PII can be stripped *before* it ever reaches the foundation model. For anyone building under Kenya's Data Protection Act 2019, that is the data-minimisation control implemented at the boundary rather than bolted on afterwards.

Automated Reasoning checks are a different category of thing. Every other policy asks whether content falls into a risky category. This one asks whether the response actually follows the rules. You supply a source policy document, Bedrock extracts formal logic from it, you test and refine, and an immutable version attaches to the guardrail. It uses formal verification rather than sampling — AWS reports up to 99% accuracy in identifying correct responses.

Classic and Standard tiers

Content filters and denied topics run in one of two tiers. Classic is the original engine. Standard adds materially better recall, support for roughly 60 languages, and detection of harmful content inside code elements — comments, variable and function names, string literals — plus prompt-leakage detection. If you are building a coding assistant or anything multilingual, Standard is the one you want.

DEMO NOTE

Attendees who leave the default tier selected and then test a code-injection prompt will see nothing fire, and will conclude the guardrail is broken. Point at the tier selector explicitly during the walkthrough.

Where it plugs in

- Direct model calls — **InvokeModel** and **Converse**
- Bedrock Agents, Knowledge Bases and Flows
- Agent frameworks including Strands Agents and Bedrock AgentCore
- Anything at all, via **ApplyGuardrail**

Enforcement. The **bedrock:GuardrailIdentifier** IAM condition key lets you require a specific guardrail on Invoke and Converse calls. If the caller does not supply the matching guardrail, the request is rejected. This is what turns a guardrail from a suggestion a developer might forget into an organisational control.

Cost model

Billed per 1,000 text units, per enabled policy. Content filters and denied topics are \$0.15 per 1,000 text units each; Automated Reasoning checks are \$0.17. A guardrail with two policies enabled bills for two policies — the ones you leave switched off cost nothing. Check the Bedrock pricing page for current figures before quoting them.

Regional notes

Bedrock has been available in Africa (Cape Town), af-south-1, since November 2025. Global cross-Region inference from af-south-1 reaches models not natively hosted there while keeping CloudWatch and CloudTrail logs in-Region, which matters when your data-residency story has to survive an audit.

PLAN AROUND THIS

Automated Reasoning checks are available only in N. Virginia, Ohio, Oregon, Frankfurt, Ireland, Paris and Sydney — not af-south-1. They are English-only, do not support streaming APIs, and CloudFormation support is still pending.

If the session covers Automated Reasoning, tell participants which Region to use in the prerequisites message. Otherwise the first stretch of the workshop goes to debugging people who defaulted to Cape Town.

What Guardrails does not do

Worth saying out loud, because it is the most common misunderstanding:

- **It is not authentication or authorisation.** It does not know who the user is or what they are permitted to see.
- **It evaluates content, not actions.** If your agent holds a tool that can delete records, a guardrail on the text will not stop the tool call.
- **It is not a substitute for input validation** in your own application layer.
- **It reduces risk probabilistically** for most policy types. Only Automated Reasoning checks offer formal guarantees, and only against the specific policy you supplied.

Guardrails are one layer. Treat it as the content boundary in a defence-in-depth design, not as the whole security posture.

Sources

Bedrock Guardrails product page aws.amazon.com/bedrock/guardrails/

Guardrails documentation docs.aws.amazon.com/bedrock/latest/userguide/guardrails.html

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